SANT'ANNA WORKSHOP ON CAUSAL INFERENCE

Covariates in DiD

OLS specs, IPW, OR, DR

Scott Cunningham · Baylor University

Sant'Anna Workshop on Causal Inference · Pisa

Recall: the 2×2 is DiD — six equivalent forms

1. $\underbrace{\big[\mathbb{E}[Y|D{=}1,t] - \mathbb{E}[Y|D{=}1,t{-}1]\big]}_{\text{first difference}} - \underbrace{\big[\mathbb{E}[Y|D{=}0,t] - \mathbb{E}[Y|D{=}0,t{-}1]\big]}_{\text{first difference}}$
2. $\underbrace{\big[\mathbb{E}[Y|D{=}1,t] - \mathbb{E}[Y|D{=}0,t]\big]}_{\text{group difference}} - \underbrace{\big[\mathbb{E}[Y|D{=}1,t{-}1] - \mathbb{E}[Y|D{=}0,t{-}1]\big]}_{\text{group difference}}$
3. $Y = \alpha_1 + \alpha_2\,\mathrm{Post} + \alpha_3\,D + \delta\,(\mathrm{Post} \times D) + \varepsilon$ *saturated regression*
4. $Y = \alpha_1 + \tau_t + \sigma_i + \delta\,(\mathrm{Post} \times D) + \varepsilon$ *two-way fixed effects*
5. $\Delta Y_i = \alpha_1 + \delta\,D_i + \varepsilon_i$ *first-difference / horizontal regression*
6. $\Delta \bar{Y}_G = \alpha_1 + \delta\,\mathrm{Post}_G + \varepsilon_G$ *group / vertical regression*

All six are **numerically identical**.

The 2×2 DiD estimand

$$\underbrace{\widehat{\delta}_{\text{DiD}}}_{\text{what we estimate}} = \underbrace{\text{ATT}}_{\text{what we want}} + \underbrace{\Big[\mathbb{E}[\Delta Y(0) \mid D{=}1] - \mathbb{E}[\Delta Y(0) \mid D{=}0]\Big]}_{\text{PT bias}}$$

Treatment = college education. Outcome = wages.
Parallel trends requires the PT bias term to equal zero.

A thought experiment

Group	Untreated wage trend $\Delta Y(0)$
Male workers	$+10$ per year
Female workers	$+8$ per year

Case 1: $D=1$ is 75% male. $D=0$ is 75% male.

Will parallel trends hold?

Case 1: show the work

	Calculation	**Result**
$\mathbb{E}[\Delta Y(0) \mid D{=}1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D{=}0]$	$0.75 \times 10 + 0.25 \times 8$	9.5

Parallel trends holds? **Yes.**

$\widehat{\delta}_{\text{DiD}} = \text{ATT} \ \checkmark$

Case 2: unequal composition

	% male
$D = 1$ (college educated)	75%
$D = 0$ (no college)	25%

Same wage trends as before ($+10$ male, $+8$ female).
Will parallel trends hold now?

Case 2: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D{=}1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D{=}0]$	$0.25 \times 10 + 0.75 \times 8$	8.5

Parallel trends holds? **No.**

$\widehat{\delta}_{\text{DiD}} \neq$ ATT because PT bias $= 9.5 - 8.5 = 1.0 \neq 0$

Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same $Y(0)$ *trends*.

Imbalance in a covariate that *causes* $Y(0)$ trends
will **mechanically** break parallel trends.

Fix: control for sex

If sex is the only covariate driving differential trends,
then *conditioning on sex* removes the PT bias.

Several ways to do this:

1. TWFE saturated regressions
2. Inverse probability weighting (IPW)
3. Outcome regression (OR)
4. Doubly robust (DR)

We will discuss all four.

For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same* $Y(0)$ trend as male control workers
2. Female treated workers have the *same* $Y(0)$ trend as female control workers

Conditional PTA: $\mathbb{E}[\Delta Y(0) \mid D{=}1,\ X] = \mathbb{E}[\Delta Y(0) \mid D{=}0,\ X]$ for all X

The standard TWFE with additive X

$$Y_{it} \;=\; \alpha_i + \lambda_t + \delta\,(D_i \times \mathrm{Post}_t) + \boldsymbol{\beta}\,X_i + \varepsilon_{it}$$

Two hidden assumptions:

1. Returns to X are **constant over time**
2. Treatment effect is **the same for all X**

Fatal problem: if X_i is time-invariant, it is collinear with α_i and drops out — whether you use unit fixed effects *or* a saturated dummy specification. An additive, separable X cannot deliver conditional parallel trends if the covariate is absorbed before it can do any work.

What if returns to X change over time?

Example: discrimination against women is not constant — $\beta_{\text{post}} \neq \beta_{\text{pre}}$.

Allow time-varying returns in the DGP:

$$Y_{it}(0) = \alpha_i + \gamma_t + \beta_t X_i$$

Write out the **post** and **pre** equations for unit i:

$$Y_{i,\text{post}}(0) = \alpha_i + \gamma_{\text{post}} + \beta_{\text{post}} X_i$$

$$Y_{i,\text{pre}}(0) = \alpha_i + \gamma_{\text{pre}} + \beta_{\text{pre}} X_i$$

Subtract (post $-$ pre):

$$\Delta Y_i(0) = \underbrace{(\gamma_{\text{post}} - \gamma_{\text{pre}})}_{\Delta\gamma} + \underbrace{(\beta_{\text{post}} - \beta_{\text{pre}})}_{\Delta\beta} \cdot X_i$$

Now take expectations by group

Starting from $\Delta Y_i(0) = \Delta\gamma + \Delta\beta \cdot X_i$, condition on treatment status:

$$E[\Delta Y(0) \mid D{=}1] = \Delta\gamma + \Delta\beta \cdot E[X \mid D{=}1]$$
$$E[\Delta Y(0) \mid D{=}0] = \Delta\gamma + \Delta\beta \cdot E[X \mid D{=}0]$$

Parallel trends bias:

$$\underbrace{E[\Delta Y(0) \mid D{=}1] - E[\Delta Y(0) \mid D{=}0]}_{\text{PT bias}} \;=\; \Delta\beta \;\cdot\; \big(\, E[X \mid D{=}1] - E[X \mid D{=}0] \,\big)$$

PT fails whenever **both**: returns shift ($\Delta\beta \neq 0$) **and** groups are imbalanced on X.

Worry 1: β changes at the policy break

$$Y_{it} \;=\; \alpha_i + \lambda_t + \delta\,(D_i \times \mathrm{Post}_t) + \boldsymbol{\beta}\,X_i + \boldsymbol{\gamma}\,(\mathrm{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post.

Worry 2: treatment effect varies with X

$$Y_{it} \; = \; \alpha_i + \lambda_t + \delta_0 \, (D_i \times \text{Post}_t) + \boldsymbol{\tau} \, (D_i \times \text{Post}_t \times X_i) + \boldsymbol{\beta} \, X_i + \varepsilon_{it}$$

$$\text{ATT} = \tfrac{1}{n_T} \sum_{D_i = 1} \left(\hat{\delta}_0 + \hat{\boldsymbol{\tau}} \, X_i \right)$$

Both worries: the fully saturated spec

$$Y_{it} = \alpha_i + \lambda_t + \delta_0\,(D_i \times \mathrm{Post}_t) + \boldsymbol{\tau}\,(D_i \times \mathrm{Post}_t \times X_i) + \boldsymbol{\gamma}\,(\mathrm{Post}_t \times X_i) + \boldsymbol{\beta}\,X_i + \varepsilon_{it}$$

$\text{ATT} = \frac{1}{n_T} \sum_{D_i=1} \left(\hat{\delta}_0 + \hat{\boldsymbol{\tau}}\,X_i\right)$ · This is **Spec C** in the lab.

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^{n} \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \, \Delta Y_i$$

$\hat{p}(X_i) = \widehat{\Pr}(D_i = 1 \mid X_i)$ (propensity score) $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$

IPW: five steps

1. **Estimate $\hat{p}(X)$.** Logit of D_i on X_i.
2. **Check overlap.** Plot $\hat{p}$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i))/(1 - \hat{p}(X_i))$.
4. **Weighted average.** $\widehat{\mathrm{ATT}} = \sum w_i \, \Delta Y_i \; / \; n_T$.
5. **Standard error.** Bootstrap or influence function.

Overlap is non-negotiable

If $\hat{p}(X_i) \to 1$ for a control unit, its weight $\to \infty$.

No overlap = no valid counterfactual.

This is a data problem. No estimator fixes it.

OR: model the trend, impute the counterfactual

$$\widehat{\mathrm{ATT}}_{\mathrm{OR}} = \frac{1}{n_T} \sum_{D_i=1} \Big[\Delta Y_i - \hat{\mu}_0(X_i) \Big]$$

$\hat{\mu}_0(X_i) = \widehat{E}[\Delta Y \mid D = 0,\ X]$ estimated on controls only

OR: five steps

1. **First-difference.** $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.
2. **Fit outcome model.** Regress ΔY on X *using controls only*.
3. **Predict counterfactual.** Evaluate $\hat{\mu}_0(X_i)$ for each treated unit.
4. **Mean gap.** ATT $= \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

DR: two chances to be right

$$\widehat{\mathrm{ATT}}_{\mathrm{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^{n} \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i)\right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
✓	✓	✓
✓	×	✓
×	✓	✓
×	×	**No**

"Doubly robust" $\neq$ doubly unbeatable.

Lab: ten estimators, one benchmark

`Labs/Lalonde-Covariates/lalonde_all_specs.do`

NSW treated + CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **$1,794**

Spec 0 (Naive) · Spec A (Additive X) · Spec B (Post$\times X$) · Spec C (Saturated)

OR by hand · OR `drdid` · IPW by hand · IPW `drdid` · DR by hand · DR `drdid`

A Monte Carlo to rank the estimators

Estimators:	TWFE, OR, IPW, DR
DGPs:	4 (vary whether OR model and pscore are correct)
n:	1,000 Replications: 10,000
Report:	Bias, RMSE, SE, coverage, CI length

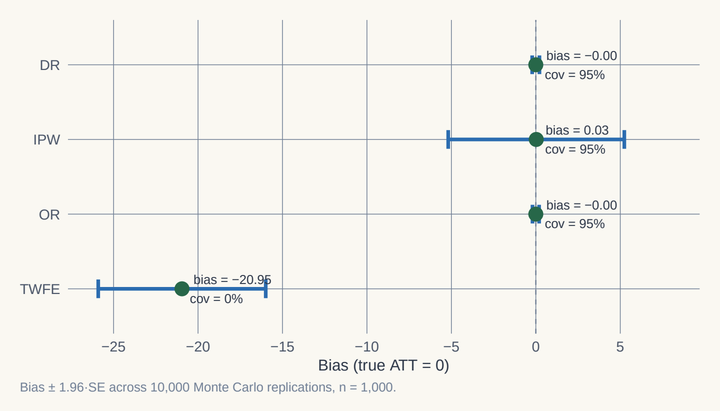
When does DR pay off? When does it break?

DGP 1: both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	-20.95	21.12	2.53	0.000	9.91
OR	-0.001	0.10	0.10	0.950	0.40
IPW	$+0.026$	2.77	2.66	0.952	10.44
DR	-0.001	0.11	0.11	0.947	0.41

TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.

DGP 1: the sampling distribution

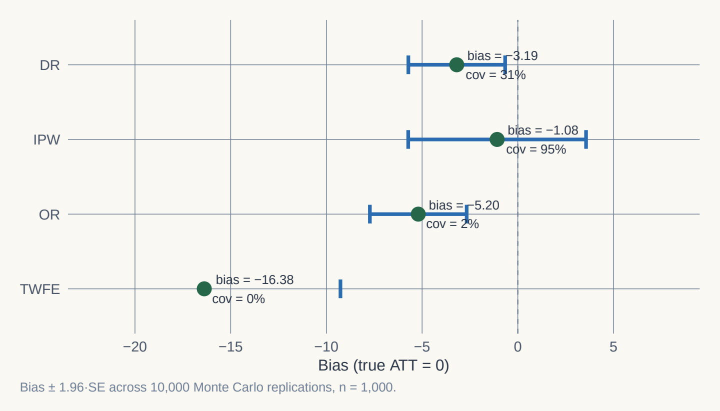


DGP 4: both models wrong

	Bias	RMSE	SE	Coverage	CI length
TWFE	-16.38	16.54	3.63	0.000	14.22
OR	-5.20	5.36	1.29	0.015	5.05
IPW	-1.08	2.66	2.37	0.949	9.31
DR	-3.19	3.45	1.29	0.308	5.07

DR undercovers at 0.31. IPW wins. Doubly robust $\neq$ doubly unbeatable.

DGP 4: the sampling distribution



What the Monte Carlo teaches

- **TWFE + additive X:** coverage zero in every DGP. **Stop using it as a default.**
- **OR:** efficient when right. Coverage 1.5% when wrong.
- **IPW:** less efficient, but robust to outcome misspecification.
- **DR:** best when one model is right. No better than IPW when both are wrong.

Adding X doesn't make your DiD credible

Adding X shifts the credibility question—

from **"does PT hold?"**

to **"is my model of $Y(0)$ or $p(X)$ correct?"**

You still have to answer it.

Now let's see it with real data

LaLonde (1986) · Dehejia & Wahba (2002)

A famous dataset where we **know the true effect.**

The benchmark: **$1,794**.

LaLonde (1986): the RCT found ≈\$800

- Job training program (National Supported Work Demonstration)
- Randomized — treated and control groups assigned by lottery
- RCT estimate: earnings effect ≈ **\$800**

A clean causal estimate. The benchmark exists.

LaLonde: dropped the RCT controls, used CPS instead

- Replaced the randomized controls with observational data (CPS, PSID)
- Re-ran every evaluator method from the literature
- Results ranged from $-$**\$16,000 to** $+$**\$3,000**

The true effect was $800. Not one method recovered it.
"The available evidence calls into serious question the ability of nonexperimental evaluators. . . "

Empirical labor economics faces an evaluation problem.

Dehejia & Wahba (2002): propensity scores get close

- Need **two pre-treatment years** for all datasets — uses a shorter sample
- RCT benchmark on this shorter sample: **$1,794**
- Applied propensity score matching on the non-experimental sample
- Result: estimates close to $1,794

Selective use of the data + right method = nearly recovered the truth.

Our exercise: DiD with and without covariates

NSW treated + CPS controls · Pre = 1975 · Post = 1978

We know the effect: **$1,794**

Let's see which specs recover it — and which don't.

Naive TWFE: $3,621 — almost double the truth

```
reg re i.post##i.ever_treated, robust
```

Estimator	ATT	vs benchmark
Naive TWFE	**$3,621**	+$1,827

No covariates — the CPS controls trend very differently from NSW treated.

Additive X: still \$3,621

```
reg re i.post##i.ever_treated $X, robust
```

Estimator	ATT	vs benchmark
Additive X	**\$3,621**	$+$\$1,827

Adding X as time-invariant regressors does nothing here. The unit FE absorbs the level differences. The trend bias remains.

Post $\times X$: \$1,711 — now in the right range

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree ///
    i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

Estimator	ATT	vs benchmark
Post $\times X$	**\$1,711**	$-$\$83

Saturated TWFE: $1,770

```
* First-difference, then regress dy on X + D x X interactions
reg dy $X ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.re74 ///
    i.ever_treated, robust
* Predict counterfactuals; ATT = mean(tau_sat) for treated
```

Estimator	ATT	vs benchmark
Saturated TWFE	**$1,770**	−$24

OR (HIT 1997): same answer as saturated TWFE

```
* Fit trend model on controls only
reg dy $X if ever_treated == 0, robust
predict dy_hat
* ATT = mean gap for treated
gen delta_hat = dy - dy_hat if ever_treated == 1
```

Estimator	ATT	vs benchmark
OR by hand / `drdid reg`	**$1,770**	−$24

Saturated TWFE and OR are algebraically equivalent in the 2-period case.

IPW (Abadie 2005): $1,861

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

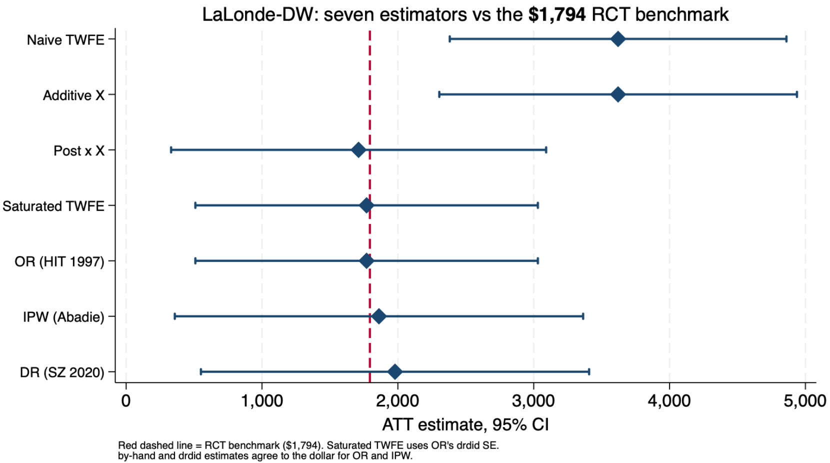
Estimator	ATT	vs benchmark
IPW	**$1,861**	+$67

DR (Sant'Anna–Zhao 2020): $1,979

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Estimator	ATT	vs benchmark
DR	**$1,979**	+$185

All seven estimators vs the $1,794 benchmark



Covariates are not robustness checks — they are design

- People say: *“if adding covariates changes results, be suspicious.”*
- That gets it backwards.
- We **know** the effect is $1,794.
- Without covariates, or with additive covariates, we get $3,621 — *wrong*.
- The correct specs recover $1,770–$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

The question is always

"did I get the model right?"